

FLI1 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes FLI1 gene fusions from the msk_impact_50k_2026 study, covering 118 fusion events. 73/117 fusions (62.4%) were in-frame. 110/117 fusions (94.0%) retained the ETS DNA-binding domain. Domain retention was tested with Fisher's exact test ($p=0.0687101$, not statistically significant at $\alpha=0.05$). No literature benchmark is configured for FLI1.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 118 analyzed fusion events, identifying 3 distinct fusion partner genes. The most frequent partner was EWSR1, observed in 116 events.

Domain retention

ETS DNA-binding domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 71/73 (97.3%) of in-frame fusions retained the domain, $p=0.0687101$ (not statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.453546.

Domain disruption

Domain-disruption analysis was skipped: FLI1 has no disruption_required_domains configured; domain-disruption analysis was skipped.

Cutpoint detection

Cutpoint detection scanned 117 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 241 aa (exon 6/7 boundary; permutation-corrected $p=0.004995$, statistically significant at $\alpha=0.05$). This is -39 aa from the nearest configured domain boundary at 280 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" ($n=110$) and "not_retained" ($n=7$). For Frame_status, retained: 71/102 (69.6%, 95% CI 60.1%-77.7%); not_retained: 2/5 (40.0%, 95% CI 11.8%-76.9%). Welch's t-test comparing Tumor_variant_count between groups gave means of 66.9091 vs 20.7143, $p=2.40299e-05$ (statistically significant at $\alpha=0.05$).

Composite evidence score

Composite evidence score reported: domain_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption_required_domains not configured for this gene).

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Expression association

Expression association reported: No mRNA expression data was available for FLI1 in this cohort; expression-association analysis was skipped.

Joint partner

Joint partner reported: FLI1 has no gene_pair configured; joint-partner analysis was skipped.

Mutation cooccurrence

Mutation cooccurrence reported: FLI1 has no mutual_exclusivity_targets configured; co-occurrence/mutual-exclusivity analysis was skipped.

Window detection

Window detection produced results for this run (35 row(s) in window_scan, 17 row(s) in top_windows); see the accompanying table.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
EWSR1	116	0.9830508474576272	0.03433782246222147	None	0.7245283018867925	0.5862586106414931	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.5885060202500145	1
ATM	1	0.00847457627118644	0.03433782246222147	None	0.8113207547169812	0.5862586106414931	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.2393135034607597	2
ERG	1	0.00847457627118644	0.03433782246222147	None	0.0	0.5862586106414931	confidence_certainty, cutpoint_proximity, domain_retention, recurrence	0.08719086195132575	3

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
129	3	107	0	7	None	1.0	-0.0
197	29	81	1	6	2.1481481481481484	0.6759093008029847	0.1701115774363377
219	105	5	4	3	15.75	0.006584775670191042	2.1814590159729152
241	107	3	4	3	26.75	0.0024830075378275904	2.6050219620274095
261	109	1	6	1	18.166666666666668	0.11656351311523727	0.9334373718114223
277	110	0	6	1	None	0.059829059829059825	1.2230878217319048

domain_retention tables

frame_domain_contingency_table

71	39
2	5

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
ets	ETS DNA-binding domain	117	110	1	6	0.11703958691910499

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
9	110	118	0.9322033898305084

frequency tables

Partner_gene_counts

Partner_gene	Event_count
ATM	1
ERG	1
EWSR1	116

window_detection tables

window_scan

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
-3	197	200	30	87	29	81	2.1481481481481484	0.6759093008029847	0.1701115774363377	EVT-P-0001233-T01-IM3-119, EVT-P-0002240-T01-IM3-6, EVT-P-0003290-T01-IM5-8, EVT-P-0004931-T02-IM5-11, EVT-P-0008030-T01-IM5-18, ... (30 total)
19	219	200	109	8	105	5	15.75	0.006584775670191042	2.1814590159729152	EVT-P-0000253-T02-IM5-2, EVT-P-0000719-T01-IM3-51, EVT-P-0001233-T01-IM3-119, EVT-P-0001377-T01-IM3-4, EVT-P-0001896-T01-IM3-5, ... (109 total)
41	241	200	111	6	107	3	26.75	0.0024830075378275904	2.6050219620274095	EVT-P-0000253-T02-IM5-2, EVT-P-0000719-T01-IM3-51, EVT-P-0001233-T01-IM3-119, EVT-P-0001377-T01-IM3-4, EVT-P-0001896-T01-IM3-5, ... (111 total)
97	197	100	30	87	29	81	2.1481481481481484	0.6759093008029847	0.1701115774363377	EVT-P-0001233-T01-IM3-119, EVT-P-0002240-T01-IM3-6, EVT-P-0003290-T01-IM5-8, EVT-P-0004931-T02-IM5-11, EVT-P-0008030-T01-IM5-18, ... (30 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
119	219	100	109	8	105	5	15.75	0.006584775670191042	2.1814590159729152	EVT-P-0000253-T02-IM5-2, EVT-P-0000719-T01-IM3-51, EVT-P-0001233-T01-IM3-119, EVT-P-0001377-T01-IM3-4, EVT-P-0001896-T01-IM3-5, ... (109 total)
129	229	100	109	8	105	5	15.75	0.006584775670191042	2.1814590159729152	EVT-P-0000253-T02-IM5-2, EVT-P-0000719-T01-IM3-51, EVT-P-0001233-T01-IM3-119, EVT-P-0001377-T01-IM3-4, EVT-P-0001896-T01-IM3-5, ... (109 total)
141	241	100	108	9	104	6	13.0	0.00961039325753103	2.017258840626846	EVT-P-0000253-T02-IM5-2, EVT-P-0000719-T01-IM3-51, EVT-P-0001233-T01-IM3-119, EVT-P-0001377-T01-IM3-4, EVT-P-0001896-T01-IM3-5, ... (108 total)
147	197	50	27	90	26	84	1.8571428571428572	1.0	-0.0	EVT-P-0001233-T01-IM3-119, EVT-P-0004931-T02-IM5-11, EVT-P-0008030-T01-IM5-18, EVT-P-0008102-T01-IM5-19, EVT-P-0008731-T01-IM5-23, ... (27 total)
161	261	100	112	5	106	4	4.416666666666667	0.26952123522631927	0.5694070116338482	EVT-P-0000253-T02-IM5-2, EVT-P-0000719-T01-IM3-51, EVT-P-0001233-T01-IM3-119, EVT-P-0001377-T01-IM3-4, EVT-P-0001896-T01-IM3-5, ... (112 total)
169	219	50	106	11	102	8	9.5625	0.017863852537282014	1.748024874940483	EVT-P-0000253-T02-IM5-2, EVT-P-0000719-T01-IM3-51, EVT-P-0001233-T01-IM3-119, EVT-P-0001377-T01-IM3-4, EVT-P-0001896-T01-IM3-5, ... (106 total)
172	197	25	27	90	26	84	1.8571428571428572	1.0	-0.0	EVT-P-0001233-T01-IM3-119, EVT-P-0004931-T02-IM5-11, EVT-P-0008030-T01-IM5-18, EVT-P-0008102-T01-IM5-19, EVT-P-0008731-T01-IM5-23, ... (27 total)
177	277	100	113	4	107	3	5.944444444444445	0.22128207151484985	0.6550537716018223	EVT-P-0000253-T02-IM5-2, EVT-P-0000719-T01-IM3-51, EVT-P-0001233-T01-IM3-119, EVT-P-0001377-T01-IM3-4, EVT-P-0001896-T01-IM3-5, ... (113 total)
191	241	50	108	9	104	6	13.0	0.00961039325753103	2.017258840626846	EVT-P-0000253-T02-IM5-2, EVT-P-0000719-T01-IM3-51, EVT-P-0001233-T01-IM3-119, EVT-P-0001377-T01-IM3-4, EVT-P-0001896-T01-IM3-5, ... (108 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
194	219	25	106	11	102	8	9.5625	0.017863852537282014	1.748024874940483	EVT-P-0000253-T02-IM5-2, EVT-P-0000719-T01-IM3-51, EVT-P-0001233-T01-IM3-119, EVT-P-0001377-T01-IM3-4, EVT-P-0001896-T01-IM3-5, ... (106 total)
197	222	25	106	11	102	8	9.5625	0.017863852537282014	1.748024874940483	EVT-P-0000253-T02-IM5-2, EVT-P-0000719-T01-IM3-51, EVT-P-0001233-T01-IM3-119, EVT-P-0001377-T01-IM3-4, EVT-P-0001896-T01-IM3-5, ... (106 total)
197	247	50	108	9	104	6	13.0	0.00961039325753103	2.017258840626846	EVT-P-0000253-T02-IM5-2, EVT-P-0000719-T01-IM3-51, EVT-P-0001233-T01-IM3-119, EVT-P-0001377-T01-IM3-4, EVT-P-0001896-T01-IM3-5, ... (108 total)
197	297	100	113	4	107	3	5.944444444444445	0.22128207151484985	0.6550537716018223	EVT-P-0000253-T02-IM5-2, EVT-P-0000719-T01-IM3-51, EVT-P-0001233-T01-IM3-119, EVT-P-0001377-T01-IM3-4, EVT-P-0001896-T01-IM3-5, ... (113 total)
211	261	50	85	32	80	30	1.0666666666666667	1.0	-0.0	EVT-P-0000253-T02-IM5-2, EVT-P-0000719-T01-IM3-51, EVT-P-0001377-T01-IM3-4, EVT-P-0001896-T01-IM3-5, EVT-P-0003272-T01-IM5-7, ... (85 total)
216	241	25	81	36	78	32	3.25	0.19942776003078416	0.7002143887071214	EVT-P-0000253-T02-IM5-2, EVT-P-0000719-T01-IM3-51, EVT-P-0001377-T01-IM3-4, EVT-P-0001896-T01-IM3-5, EVT-P-0003272-T01-IM5-7, ... (81 total)
219	244	25	81	36	78	32	3.25	0.19942776003078416	0.7002143887071214	EVT-P-0000253-T02-IM5-2, EVT-P-0000719-T01-IM3-51, EVT-P-0001377-T01-IM3-4, EVT-P-0001896-T01-IM3-5, EVT-P-0003272-T01-IM5-7, ... (81 total)
219	269	50	85	32	80	30	1.0666666666666667	1.0	-0.0	EVT-P-0000253-T02-IM5-2, EVT-P-0000719-T01-IM3-51, EVT-P-0001377-T01-IM3-4, EVT-P-0001896-T01-IM3-5, EVT-P-0003272-T01-IM5-7, ... (85 total)
219	319	100	86	31	81	29	1.1172413793103448	1.0	-0.0	EVT-P-0000253-T02-IM5-2, EVT-P-0000719-T01-IM3-51, EVT-P-0001377-T01-IM3-4, EVT-P-0001896-T01-IM3-5, EVT-P-0003272-T01-IM5-7, ... (86 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
219	419	200	87	30	81	29	0.46551724137931033	0.6759093008029846	0.17011157743633779	EVT-P-0000253-T02-IM5-2, EVT-P-0000719-T01-IM3-51, EVT-P-0001377-T01-IM3-4, EVT-P-0001896-T01-IM3-5, EVT-P-0003272-T01-IM5-7, ... (87 total)
227	277	50	7	110	5	105	0.11904761904761904	0.056053623223199824	1.2513963100094996	EVT-P-0013874-T01-IM5-31, EVT-P-0019756-T01-IM6-46, EVT-P-0024080-T02-IM6-1, EVT-P-0030220-T01-IM6-60, EVT-P-0032382-T01-IM6-64, ... (7 total)
236	261	25	6	111	4	106	0.09433962264150944	0.041246621234544124	1.3846116213999358	EVT-P-0013874-T01-IM5-31, EVT-P-0019756-T01-IM6-46, EVT-P-0024080-T02-IM6-1, EVT-P-0030220-T01-IM6-60, EVT-P-0032382-T01-IM6-64, ... (6 total)
241	266	25	6	111	4	106	0.09433962264150944	0.041246621234544124	1.3846116213999358	EVT-P-0013874-T01-IM5-31, EVT-P-0019756-T01-IM6-46, EVT-P-0024080-T02-IM6-1, EVT-P-0030220-T01-IM6-60, EVT-P-0032382-T01-IM6-64, ... (6 total)
241	291	50	7	110	5	105	0.11904761904761904	0.056053623223199824	1.2513963100094996	EVT-P-0013874-T01-IM5-31, EVT-P-0019756-T01-IM6-46, EVT-P-0024080-T02-IM6-1, EVT-P-0030220-T01-IM6-60, EVT-P-0032382-T01-IM6-64, ... (7 total)
241	341	100	7	110	5	105	0.11904761904761904	0.056053623223199824	1.2513963100094996	EVT-P-0013874-T01-IM5-31, EVT-P-0019756-T01-IM6-46, EVT-P-0024080-T02-IM6-1, EVT-P-0030220-T01-IM6-60, EVT-P-0032382-T01-IM6-64, ... (7 total)
241	441	200	8	109	5	105	0.06349206349206349	0.0065847756701910435	2.1814590159729152	EVT-P-0013874-T01-IM5-31, EVT-P-0019756-T01-IM6-46, EVT-P-0024080-T02-IM6-1, EVT-P-0030220-T01-IM6-60, EVT-P-0032382-T01-IM6-64, ... (8 total)
247	347	100	6	111	3	107	0.037383177570093455	0.00248300753782759	2.6050219620274095	EVT-P-0013874-T01-IM5-31, EVT-P-0019756-T01-IM6-46, EVT-P-0024080-T02-IM6-1, EVT-P-0039688-T01-IM6-77, EVT-P-0045722-T01-IM6-88, ... (6 total)
252	277	25	5	112	3	107	0.07009345794392523	0.02832541666897195	1.5478236929136615	EVT-P-0013874-T01-IM5-31, EVT-P-0019756-T01-IM6-46, EVT-P-0024080-T02-IM6-1, EVT-P-0039688-T01-IM6-77, EVT-P-0045722-T01-IM6-88

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
261	286	25	5	112	3	107	0.07009345794392523	0.02832541666897195	1.5478236929136615	EVT-P-0013874-T01-IM5-31, EVT-P-0019756-T01-IM6-46, EVT-P-0024080-T02-IM6-1, EVT-P-0039688-T01-IM6-77, EVT-P-0045722-T01-IM6-88
261	311	50	5	112	3	107	0.07009345794392523	0.02832541666897195	1.5478236929136615	EVT-P-0013874-T01-IM5-31, EVT-P-0019756-T01-IM6-46, EVT-P-0024080-T02-IM6-1, EVT-P-0039688-T01-IM6-77, EVT-P-0045722-T01-IM6-88
261	361	100	6	111	3	107	0.037383177570093455	0.00248300753782759	2.6050219620274095	EVT-P-0013874-T01-IM5-31, EVT-P-0019756-T01-IM6-46, EVT-P-0024080-T02-IM6-1, EVT-P-0039688-T01-IM6-77, EVT-P-0045722-T01-IM6-88, ... (6 total)
261	461	200	6	111	3	107	0.037383177570093455	0.00248300753782759	2.6050219620274095	EVT-P-0013874-T01-IM5-31, EVT-P-0019756-T01-IM6-46, EVT-P-0024080-T02-IM6-1, EVT-P-0039688-T01-IM6-77, EVT-P-0045722-T01-IM6-88, ... (6 total)

top_windows

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
247	347	100	6	111	3	107	0.037383177570093455	0.00248300753782759	2.6050219620274095	EVT-P-0013874-T01-IM5-31, EVT-P-0019756-T01-IM6-46, EVT-P-0024080-T02-IM6-1, EVT-P-0039688-T01-IM6-77, EVT-P-0045722-T01-IM6-88, ... (6 total)
41	241	200	111	6	107	3	26.75	0.0024830075378275904	2.6050219620274095	EVT-P-0000253-T02-IM5-2, EVT-P-0000719-T01-IM3-51, EVT-P-0001233-T01-IM3-119, EVT-P-0001377-T01-IM3-4, EVT-P-0001896-T01-IM3-5, ... (111 total)
119	219	100	109	8	105	5	15.75	0.006584775670191042	2.1814590159729152	EVT-P-0000253-T02-IM5-2, EVT-P-0000719-T01-IM3-51, EVT-P-0001233-T01-IM3-119, EVT-P-0001377-T01-IM3-4, EVT-P-0001896-T01-IM3-5, ... (109 total)
241	441	200	8	109	5	105	0.06349206349206349	0.0065847756701910435	2.1814590159729152	EVT-P-0013874-T01-IM5-31, EVT-P-0019756-T01-IM6-46, EVT-P-0024080-T02-IM6-1, EVT-P-0030220-T01-IM6-60, EVT-P-0032382-T01-IM6-64, ... (8 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
191	241	50	108	9	104	6	13.0	0.00961039325753103	2.017258840626846	EVT-P-0000253-T02-IM5-2, EVT-P-0000719-T01-IM3-51, EVT-P-0001233-T01-IM3-119, EVT-P-0001377-T01-IM3-4, EVT-P-0001896-T01-IM3-5, ... (108 total)
194	219	25	106	11	102	8	9.5625	0.017863852537282014	1.748024874940483	EVT-P-0000253-T02-IM5-2, EVT-P-0000719-T01-IM3-51, EVT-P-0001233-T01-IM3-119, EVT-P-0001377-T01-IM3-4, EVT-P-0001896-T01-IM3-5, ... (106 total)
252	277	25	5	112	3	107	0.07009345794392523	0.02832541666897195	1.5478236929136615	EVT-P-0013874-T01-IM5-31, EVT-P-0019756-T01-IM6-46, EVT-P-0024080-T02-IM6-1, EVT-P-0039688-T01-IM6-77, EVT-P-0045722-T01-IM6-88
236	261	25	6	111	4	106	0.09433962264150944	0.041246621234544124	1.3846116213999358	EVT-P-0013874-T01-IM5-31, EVT-P-0019756-T01-IM6-46, EVT-P-0024080-T02-IM6-1, EVT-P-0030220-T01-IM6-60, EVT-P-0032382-T01-IM6-64, ... (6 total)
227	277	50	7	110	5	105	0.11904761904761904	0.056053623223199824	1.2513963100094996	EVT-P-0013874-T01-IM5-31, EVT-P-0019756-T01-IM6-46, EVT-P-0024080-T02-IM6-1, EVT-P-0030220-T01-IM6-60, EVT-P-0032382-T01-IM6-64, ... (7 total)
216	241	25	81	36	78	32	3.25	0.19942776003078416	0.7002143887071214	EVT-P-0000253-T02-IM5-2, EVT-P-0000719-T01-IM3-51, EVT-P-0001377-T01-IM3-4, EVT-P-0001896-T01-IM3-5, EVT-P-0003272-T01-IM5-7, ... (81 total)
177	277	100	113	4	107	3	5.944444444444445	0.22128207151484985	0.6550537716018223	EVT-P-0000253-T02-IM5-2, EVT-P-0000719-T01-IM3-51, EVT-P-0001233-T01-IM3-119, EVT-P-0001377-T01-IM3-4, EVT-P-0001896-T01-IM3-5, ... (113 total)
161	261	100	112	5	106	4	4.416666666666667	0.26952123522631927	0.5694070116338482	EVT-P-0000253-T02-IM5-2, EVT-P-0000719-T01-IM3-51, EVT-P-0001233-T01-IM3-119, EVT-P-0001377-T01-IM3-4, EVT-P-0001896-T01-IM3-5, ... (112 total)
219	419	200	87	30	81	29	0.46551724137931033	0.6759093008029846	0.17011157743633779	EVT-P-0000253-T02-IM5-2, EVT-P-0000719-T01-IM3-51, EVT-P-0001377-T01-IM3-4, EVT-P-0001896-T01-IM3-5, EVT-P-0003272-T01-IM5-7, ... (87 total)

start_aa	end_aa	width_aa	n_events_inside	n_events_outside	n_positive_inside	n_positive_outside	odds_ratio	p_value	neg_log10_p_value	event_ids_inside
97	197	100	30	87	29	81	2.1481481481481484	0.6759093008029847	0.1701115774363377	EVT-P-0001233-T01-IM3-119, EVT-P-0002240-T01-IM3-6, EVT-P-0003290-T01-IM5-8, EVT-P-0004931-T02-IM5-11, EVT-P-0008030-T01-IM5-18, ... (30 total)
172	197	25	27	90	26	84	1.8571428571428572	1.0	-0.0	EVT-P-0001233-T01-IM3-119, EVT-P-0004931-T02-IM5-11, EVT-P-0008030-T01-IM5-18, EVT-P-0008102-T01-IM5-19, EVT-P-0008731-T01-IM5-23, ... (27 total)
211	261	50	85	32	80	30	1.0666666666666667	1.0	-0.0	EVT-P-0000253-T02-IM5-2, EVT-P-0000719-T01-IM3-51, EVT-P-0001377-T01-IM3-4, EVT-P-0001896-T01-IM3-5, EVT-P-0003272-T01-IM5-7, ... (85 total)
219	319	100	86	31	81	29	1.1172413793103448	1.0	-0.0	EVT-P-0000253-T02-IM5-2, EVT-P-0000719-T01-IM3-51, EVT-P-0001377-T01-IM3-4, EVT-P-0001896-T01-IM3-5, EVT-P-0003272-T01-IM5-7, ... (86 total)

Per-event results (results.tsv)

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0024080-T02-IM6-1	P-0024080-T02-IM6		Prostate Cancer	PRAD	ATM-FLI1	ATM	out-of-frame	five_prime	128677281	7	261	True	260	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False		ETS DNA-binding domain		FLI1 (NM_002017) - ATM (NM_000051) rearrangement: c.781+147:FLI1_c.8269-128:A TMdup Protein Fusion: out of frame {FLI1:ATM}	NA
EVT-P-000253-T02-IM5-2	P-000253-T02-IM5		Bone Cancer	ES	EWSR1-FLI1	EWSR1	out-of-frame	three_prime	128664485	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1(NM_013986)-FLI1(NM_002017) Fusion (EWSR1 exon 9 fused to FLI1 6) : t(22;11)(22q12.2.13;11q24.3)(chr22:g.29685870::chr11:g.128664485) Protein fusion: out of frame (EWSR1-FLI1)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0047480-T03-IM6-3	P-0047480-T03-IM6		Soft Tissue Sarcoma	ESST	EWSR1-FLI1	EWSR1	in-frame	three_prime	128649430	5	197	True	197	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128649430::chr22:g.29688384) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-0001377-T01-IM3-4	P-0001377-T01-IM3		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128660474	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			None Protein fusion: in frame (EWSR1-FLI1)	NA
EVT-P-0001896-T01-IM3-5	P-0001896-T01-IM3		Bone Cancer	ES	EWSR1-FLI1	EWSR1	unknown	three_prime	128668840	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) Translocation: t(22;10)(22q12.2;q10q26.2)(chr22:g.29684628::chr10:g.128668840) Protein fusion: mid-exon (EWSR1-FLI1)	NA
EVT-P-0002240-T01-IM3-6	P-0002240-T01-IM3		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128638556	4	129	True	129	True	exact match	retained	1.0	False	ETS DNA-binding domain			NA Protein fusion: in frame (EWSR1-FLI1)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intro_nic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0003272-T01-IM5-7	P-0003272-T01-IM5		Head and Neck Cancer	ODGC	EWSR1-FLI1	EWSR1	in-frame	three_prime	128673862	6	219	True	219	True	exact match	retained	1.0	False	ETS DN A-binding domain			NA Protein fusion: in frame (EWSR1-FLI1)	NA
EVT-P-0003290-T01-IM5-8	P-0003290-T01-IM5		Bone Cancer	ES	EWSR1-FLI1	EWSR1	unknown	three_prime	128641490	4	129	True	129	True	exact match	retained	1.0	False	ETS DN A-binding domain			NA Protein fusion: mid-exon (EWSR1-FLI1)	NA
EVT-P-0003554-T01-IM5-9	P-0003554-T01-IM5		Bone Cancer	ES	EWSR1-FLI1	EWSR1	out-of-frame	three_prime	128664769	6	219	True	219	True	exact match	retained	1.0	False	ETS DN A-binding domain			EWSR1 (NM_013986) and FLI1 (NM_002017) fusion (EWSR1 exon 9 fused to FLI1 exon 6): t(22;11)(q12.2;q24.3)(chr22:g.29685775::chr11:g.128664769) Protein fusion: out of frame (EWSR1-FLI1)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0004498-T01-IM5-10	P-0004498-T01-IM5		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128674058	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) reciprocal fusion (EWSR1 exons 1-9 with FLI1 exons 6-9) : t(11;22)(11q24.3;22q12.3)(chr11:g.128674058::chr22:g.29688226) Protein fusion: in frame (EWSR1-FLI1)	NA
EVT-P-0004931-T02-IM5-11	P-0004931-T02-IM5		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128651438	5	197	True	197	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) reciprocal fusion (EWSR1 exons 1-10 with FLI1 exons 5-9) : t(2;11)(22q12.2;11q24.3)(chr22:g.29688357::chr11:g.128651438) Protein fusion: in frame (EWSR1-FLI1)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0005297-T01-IM5-12	P-0005297-T01-IM5		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128658432	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-8 with FLI1 exons 6-9): t(22;11)(22q12.2;11q24.3)(chr22:g.29684524::chr11:g.128658432) Protein fusion: in frame (EWSR1-FLI1)	NA
EVT-P-0005297-T02-IM6-13	P-0005297-T02-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128658433	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-8 with FLI1 exons 6-9): t(22;11)(22q12.2;11q24.3)(chr22:g.29684524::chr11:g.128658433) Protein Fusion: in frame {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0005614-T01-IM5-14	P-0005614-T01-IM5		Bone Cancer	ES	EWSR1-FLI1	EWSR1	out-of-frame	five_prime	128667277	5	219	True	218	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False		ETS DNA-binding domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-9 with FLI1 exons 6-9) : t(11;22)(11q24.3;22q12.2)(chr11:g.128667277::chr22:g.29684957) Protein fusion: out of frame (FLI1-EWSR1)	NA
EVT-P-0005716-T01-IM5-15	P-0005716-T01-IM5		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128662783	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-9 with FLI1 exons 6-9) : t(2;11)(q12.2;q14.3)(chr22:g.29688428::chr11:g.128662783) Protein fusion: in frame (EWSR1-FLI1)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0008030-T01-IM5-18	P-0008030-T01-IM5		Bone Cancer	ES	EWSR1-FLI1	EWSR1	out-of-frame	three_prime	128647136	5	197	True	197	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-9 with FLI1 exons 5-9): t(11;22)(11q24.3;q12.2)(chr11:g.128647136::chr22:g.29685491) Protein fusion: out of frame (EWSR1-FLI1)	NA
EVT-P-0008102-T01-IM5-19	P-0008102-T01-IM5		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128649909	5	197	True	197	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_001163287) - FLI1 (NM_002017) fusion (EWSR1 exons 1-7 fused with FLI1 exons 5-9) : t(11;22)(q24.3;q12.2)(chr11:g.128649909::chr22:g.29684432) Protein fusion: in frame (EWSR1-FLI1)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0008320-T01-IM5-20	P-0008320-T01-IM5		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128651930	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-8 fused to FLI1 exons 6-9) : t(11;22)(q24.3;q12.2)(chr11:g.128651930::chr22:g.29683853) Protein fusion: in frame (EWSR1-FLI1)	NA
EVT-P-0008352-T01-IM5-21	P-0008352-T01-IM5		Bone Cancer	ES	EWSR1-FLI1	EWSR1	out-of-frame	three_prime	128658456	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) -FLI1 (NM_002017) Fusion (EWSR1 exons 1-7 with FLI1 exons 6-9) : t(22;11)(q24.3;q12.2)(chr11:g.128658456) Protein fusion: out of frame (EWSR1-FLI1)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0008355-T01-IM5-22	P-0008355-T01-IM5		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128673264	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) Fusion (EWSR1 exons 1-7 fused with FLI1 exons 6-9): t(22;11)(q12.2;q24.3)(c11p11;q24.3) [chr11:g128673264] Protein fusion: in frame (EWSR1-FLI1)	NA
EVT-P-0008731-T01-IM5-23	P-0008731-T01-IM5		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128647421	5	197	True	197	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-8 fused with FLI1 exons 5-9): t(11;22)(p11;q12.2)(chr11:g128647421::chr22:g29684579) Protein fusion: in frame (EWSR1-FLI1)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_effect_on_frame
EVT-P-0012131-T01-IM5-25	P-0012131-T01-IM5		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128663970	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion : t(11;22)(q24.3;q12.2)(chr11:g.128663970::chr22:g.29688203) Protein fusion: in frame (EWSR1-FLI1)	NA
EVT-P-0012316-T01-IM5-26	P-0012316-T01-IM5		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128674054	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-9 fused to FLI1 exons 6-9) : t(11;22)(q24.3;q12.2)(chr11:g.128674054::chr22:g.29688203) Protein fusion: in frame (EWSR1-FLI1)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-012482-T02-IM5-27	P-0012482-T02-I-M5		Bone Cancer	ES	EWSR1-FLI1	EWSR1	out-of-frame	three_prime	128639397	4	129	True	129	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-9 fused to FLI1 exons 4-9): t(11;22)(q24.3;q12.2)(chr11:g.128639397::chr2:g.29684794) Protein fusion: out of frame (EWSR1-FLI1)	NA
EVT-P-013601-T01-IM5-28	P-0013601-T01-I-M5		Bone Cancer	ES	EWSR1-FLI1	EWSR1	out-of-frame	three_prime	128664219	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-9 fused with FLI1 exons 6-9): t(11;22)(q24.3;q12.2)(chr11:g.128664219::chr2:g.29685257) Protein fusion: out of frame (EWSR1-FLI1)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0013746-T02-IM5-29	P-0013746-T02-IM5		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128644899	5	197	True	197	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-11 fused to FLI1 exons 5-9): t(11;22)(q24.3;q12.2)(chr11:g.28644899::chr22:g.29688263) Protein fusion: in frame (EWSR1-FLI1)	NA
EVT-P-0013746-T04-IM7-30	P-0013746-T04-IM7		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128644900	5	197	True	197	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128644900::chr22:g.29688264) Protein Fusion: in frame {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0013874-T01-IM5-31	P-0013874-T01-IM5		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	five_prime	128677168	7	261	True	260	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False		ETS DNA-binding domain		EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-11 with FLI1 exons 8-9) : t(11;22)(q24.3;q12.2)(chr11:g.128677168::chr22:g.29688431) Protein fusion: in frame (FLI1-EWSR1)	NA
EVT-P-0065945-T03-IM7-32	P-0065945-T03-IM7		Bone Cancer	ES	EWSR1-FLI1	EWSR1	out-of-frame	three_prime	128656142	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128656142::chr22:g.29684998) Protein Fusion: out of frame [EWSR1:FLI1]	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0014314-T01-IM6-33	P-0014314-T01-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128663756	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-8 fused to FLI1 exons 6-9): t(11;22)(q24.3;q12.2)(chr11:g.128663756::chr2:g.29684463) Protein fusion: in frame (EWSR1-FLI1)	NA
EVT-P-0014984-T02-IM6-34	P-0014984-T02-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	out-of-frame	three_prime	128660148	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-9 with FLI1 exons 6-9): t(11;22)(q24.3;q12.2)(chr11:g.128660148::chr22:g.29684827) Protein fusion: out of frame (EWSR1-FLI1)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-015517-T01-IM6-36	P-0015517-T01-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	out-of-frame	three_prime	128653928	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-9 with FLI1 exons 6-9) : t(11;22)(11q24.3;22q12.2)(chr11:g.9685072::chr22:g.29685072) Protein fusion: out of frame (EWSR1-FLI1)	NA
EVT-P-015619-T03-IM6-37	P-0015619-T03-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	out-of-frame	three_prime	128663112	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-9 fused to FLI1 exons 6-9): t(11;22)(q24.3;q12.2)(chr11:g.128663112::chr22:g.29685920) Protein Fusion: out of frame {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0016346-T01-IM6-38	P-0016346-T01-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	12865108	5	197	True	197	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) Fusion (EWSR1 exons 1-8 fused with FLI1 exons 5-9) : t(11;22)(q24;q12.2)(chr11:g.128651008::chr22:g.29683915) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-0017027-T01-IM6-39	P-0017027-T01-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128646363	5	197	True	197	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-8 fused in-frame with FLI1 exons 5-9): t(11;22)(q24;q12.2)(chr11:g.128646363::chr22:g.29684426) Protein Fusion: in frame {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0018508-T01-IM6-42	P-0018508-T01-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128653753	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-8 fused in-frame with FLI1 exons 6-9): t(11;22)(q24.3;q12.2)(chr11:g.128653753::chr2:g.29684479) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-0018530-T01-IM6-43	P-0018530-T01-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128643225	5	197	True	197	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) Fusion (EWSR1 exons 1-8 fused with FLI1 exons 5-9): t(11;22)(q24.3;q12.2)(chr11:g.128643225::chr2:g.29684272) Protein Fusion: in frame {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0018663-T01-IM6-44	P-0018663-T01-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128658406	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986)-FLI1 (NM_002017) Fusion (EWSR1 exons 1-8 fused with FLI1 exons 6-9) : t(11;22)(q24.3;q12.2)(chr11:g.128658406::chr22:g.29684132) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-0019080-T02-IM6-45	P-0019080-T02-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128663272	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986)-FLI1 (NM_002017) fusion (EWSR1 exons 1-8 fused with FLI1 exons 6-9) : t(11;22)(q24.3;q12.2)(chr11:g.128663272::chr22:g.29683811) Protein Fusion: in frame {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0019756-T01-IM6-46	P-0019756-T01-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128678982	8	261	True	261	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-8 fused with FLI1 exons 8-9) : t(11;22)(q24;q12.2)(chr11:g.128678982::chr22:g.29683784) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-0020463-T01-IM6-47	P-0020463-T01-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128668415	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-8 fused with FLI1 exons 6-9) : t(11;22)(q24;q12.2)(chr11:g.128668415::chr22:g.29683880) Protein Fusion: in frame {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0021068-T01-IM6-48	P-0021068-T01-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128660877	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-8 fused with FLI1 exons 6-9): t(11;22)(q24.3;q12.2)(chr11:g.128660877::chr2:g.29683596) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-0021939-T01-IM6-49	P-0021939-T01-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128665058	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) and FLI1 (NM_002017) fusion (EWSR1 exon 8 fused to FLI1 exon 6): t(11;22)(q24.3;q12.2)(chr11:g.128665058::chr22:g.29683347) Protein Fusion: in frame {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_effect_on_frame
EVT-P-0022452-T01-IM6-50	P-0022452-T01-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128672031	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-11 fused to FLI1 exons 6-9): t(11;22)(q24.3;q12.2)(chr11:g.128672031::chr2:g.29688410) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-000719-T01-IM3-51	P-000719-T01-IM3		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128673072	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			NA Protein fusion: in frame (EWSR1-FLI1)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0024807-T01-IM6-52	P-0024807-T01-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128656386	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986)-FLI1 (NM_002017) fusion (EWSR1 exons 1-8 fused with FLI1 exons 6-9) : t(11;22)(q24.3;q12.2)(chr11:g.128656386::chr22:g.29684194) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-0024945-T01-IM6-53	P-0024945-T01-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128663120	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986)-FLI1 (NM_002017) fusion (EWSR1 exons 1-8 fused with FLI1 exons 6-9) : t(11;22)(q24.3;q12.2)(chr11:g.128663120::chr22:g.29683825) Protein Fusion: in frame {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0025251-T01-IM6-54	P-0025251-T01-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	unknown	five_prime	128662338	5	219	True	218	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False		ETS DNA-binding domain		FLI1 (NM_002017) - EWSR1 (NM_013986) Rearrangement: t(11;22)(q24.3;q12.2)(chr11:g.128662338::chr22:g.29683006) Protein Fusion: mid-exon (FLI1: EWSR1)	NA
EVT-P-0026404-T01-IM6-55	P-0026404-T01-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	out-of-frame	three_prime	128652868	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) and FLI1 (NM_002017) fusion (EWSR1 exons 1-8 fused with FLI1 exons 6-9): t(11;22)(q24.3;q12.2)(chr22:g.29684987::chr11:g.128652868) Protein Fusion: out of frame (EWSR1:FLI1)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0029448-T01-IM6-56	P-0029448-T01-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128647971	5	197	True	197	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-8 fused with FLI1 exons 5-9) : t(11;22)(q24.3;q12.2)(chr11:g.128647971::chr22:g.29684099) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-0029619-T02-IM6-57	P-0029619-T02-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128662958	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128662958::chr22:g.29688349) Protein Fusion: in frame {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0030044-T02-IM6-58	P-0030044-T02-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128653122	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_005243) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128653122::chr22:g.29684176) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-0030044-T03-IM7-59	P-0030044-T03-IM7		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128653122	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128653122::chr22:g.29684176) Protein Fusion: in frame {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0030220-T01-IM6-60	P-0030220-T01-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128676517	7	241	True	241	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128676517::chr22:g.29688327) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-0030279-T02-IM6-61	P-0030279-T02-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	unknown	three_prime	128662512	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128662512::chr22:g.29684687) Protein Fusion: mid-exon {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0030279-T03-IM7-62	P-0030279-T03-IM7		Bone Cancer	ES	EWSR1-FLI1	EWSR1	unknown	three_prime	128662512	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128662512::chr22:g.29684687) Protein Fusion: mid-exon {EWSR1:FLI1}	NA
EVT-P-0030349-T03-IM6-63	P-0030349-T03-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128666334	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-8 fused with FLI1 exons 6-9) : t(11;22)(q24.3;q12.2)(chr11:g.128666334::chr22:g.29683948) Protein Fusion: in frame {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0032382-T01-IM6-64	P-0032382-T01-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128676273	7	241	True	241	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-11 fused in-frame with FLI1 exons 7-9): t(11;22)(q24.3;q12.2)(chr11:g.128676273::chr22:g.29688342) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-0032610-T01-IM6-65	P-0032610-T01-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	out-of-frame	three_prime	128644868	5	197	True	197	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128644868::chr22:g.29685790) Protein Fusion: out of frame {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0034612-T01-IM6-66	P-0034612-T01-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128659530	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128659530::chr22:g.29683806) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-0034612-T02-IM6-67	P-0034612-T02-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128659530	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-8 fused to FLI1 exons 6-9): t(11;22)(q24.3;q12.2)(chr11:g.128659530::chr22:g.29683806) Protein Fusion: in frame {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0036245-T01-IM6-68	P-0036245-T01-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128658093	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128658093::chr22:g.29684449) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-0037173-T01-IM6-69	P-0037173-T01-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128670710	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128670710::chr22:g.29688297) Protein Fusion: in frame {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	tumor_type	oncotre_e_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0037267-T01-IM6-70	P-0037267-T01-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128672985	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-11 fused to FLI1 exons 6-9): t(11;22)(q24.3;q12.2)(chr11:g.128672985::chr2:g.29688402) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-0038127-T01-IM6-71	P-0038127-T01-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128646529	5	197	True	197	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-8 fused to FLI1 exons 5-9): t(11;22)(q24.3;q12.2)(chr11:g.28646529::chr22:g.29683923) Protein Fusion: in frame {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	tumor_type	oncotre_e_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intro_nic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0038570-T02-IM6-72	P-0038570-T02-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128664954	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-8 fused with FLI1 exons 6-9) : t(11;22)(q24.3;q12.2)(chr11:g.128664954::chr22:g.29683885) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-0038570-T03-IM6-73	P-0038570-T03-IM6		Soft Tissue Sarcoma	ESST	EWSR1-FLI1	EWSR1	in-frame	three_prime	128664954	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128664954::chr22:g.29683885) Protein Fusion: in frame {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_effect_on_frame
EVT-P-0039202-T02-IM7-74	P-0039202-T02-IM7		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128670148	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128670148::chr22:g.29688434) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-0039406-T01-IM6-75	P-0039406-T01-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	out-of-frame	three_prime	128652054	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) rearrangement: t(11;22)(q24.3;q12.2)(chr11:g.128652054::chr22:g.29685781) Protein Fusion: out of frame {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_effect_on_frame
EVT-P-0039579-T01-IM6-76	P-0039579-T01-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128665386	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) rearrangement: t(11;22)(q24.3;q12.2)(c11:g.128665386::chr22:g.29684257) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-0039688-T01-IM6-77	P-0039688-T01-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128678290	8	261	True	261	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128678290::chr22:g.29684082) Protein Fusion: in frame {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_effect_on_frame
EVT-P-0040442-T01-IM6-78	P-0040442-T01-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	out-of-frame	three_prime	128673571	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128673571::chr22:g.29685161) Protein Fusion: out of frame {EWSR1:FLI1}	NA
EVT-P-0040442-T02-IM6-79	P-0040442-T02-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	out-of-frame	three_prime	128673571	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128673571::chr22:g.29685161) Protein Fusion: out of frame {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0041943-T01-IM6-80	P-0041943-T01-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128644404	5	197	True	197	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128644404::chr22:g.29683564) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-0042913-T01-IM6-81	P-0042913-T01-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128664632	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128664632::chr22:g.29684449) Protein Fusion: in frame {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0043526-T01-IM6-82	P-0043526-T01-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128660704	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128660704::chr22:g.29684397) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-0043526-T03-IM6-83	P-0043526-T03-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128660704	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128660704::chr22:g.29684397) Protein Fusion: in frame {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_effect_on_frame
EVT-P-0044551-T01-IM6-84	P-0044551-T01-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128655556	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128655556::chr22:g.29683586) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-0044896-T02-IM6-85	P-0044896-T02-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	out-of-frame	three_prime	128658358	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128658358::chr22:g.29685203) Protein Fusion: out of frame {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0045227-T01-IM6-86	P-0045227-T01-IM6		Salivary Gland Cancer	SACA	EWSR1-FLI1	EWSR1	in-frame	three_prime	128647321	5	197	True	197	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128647321::chr22:g.29684323) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-0045227-T03-IM6-87	P-0045227-T03-IM6		Salivary Gland Cancer	OSACA	EWSR1-FLI1	EWSR1	in-frame	three_prime	128647321	5	197	True	197	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128647321::chr22:g.29684323) Protein Fusion: in frame {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0045722-T01-IM6-88	P-0045722-T01-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128679847	9	277	True	277	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128679847::chr22:g.29683765) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-0046693-T01-IM6-89	P-0046693-T01-IM6		Pancreatic Cancer	PANET	EWSR1-FLI1	EWSR1	in-frame	three_prime	128656732	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128656732::chr22:g.29683940) Protein Fusion: in frame {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0047109-T02-IM6-90	P-0047109-T02-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128666244	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128666244::chr22:g.29683718) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-0013922-T01-IM5-91	P-0013922-T01-IM5		Bone Cancer	ES	EWSR1-FLI1	EWSR1	out-of-frame	three_prime	128644690	5	197	True	197	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion (EWSR1 exons 1-10 fused to FLI1 exons 5-9): t(11;22)(q24.3;q12.2)(chr11:g.128644690::chr22:g.29688612) Protein fusion: out of frame (EWSR1-FLI1)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_effect_on_frame
EVT-P-0048417-T02-IM6-92	P-0048417-T02-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	out-of-frame	three_prime	128662931	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128662931::chr22:g.29685071) Protein Fusion: out of frame {EWSR1:FLI1}	NA
EVT-P-0068585-T02-IM7-93	P-0068585-T02-IM7		Soft Tissue Sarcoma	ESST	EWSR1-FLI1	EWSR1	in-frame	three_prime	128672830	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128672830::chr22:g.29683515) Protein Fusion: in frame {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0048577-T03-IM6-94	P-0048577-T03-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128669947	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128669947::chr22:g.29683727) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-0051333-T01-IM6-95	P-0051333-T01-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128667777	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128667777::chr22:g.29684190) Protein Fusion: in frame {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0051569-T01-IM6-96	P-0051569-T01-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	out-of-frame	three_prime	128643608	5	197	True	197	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128643608::chr22:g.29684917) Protein Fusion: out of frame {EWSR1:FLI1}	NA
EVT-P-0052277-T01-IM6-97	P-0052277-T01-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	out-of-frame	three_prime	128649484	5	197	True	197	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128649484::chr22:g.29685235) Protein Fusion: out of frame {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0052635-T01-IM6-98	P-0052635-T01-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	out-of-frame	three_prime	128660258	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128660258::chr22:g.29685095) Protein Fusion: out of frame {EWSR1:FLI1}	NA
EVT-P-0052745-T02-IM6-99	P-0052745-T02-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	unknown	three_prime	128670137	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128670137::chr22:g.29688551) Protein Fusion: mid-exon {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_effect_on_frame
EVT-P-0066379-T02-IM7-100	P-0066379-T02-IM7		Bone Cancer	ES	EWSR1-FLI1	EWSR1	out-of-frame	three_prime	128643844	5	197	True	197	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128643844::chr22:g.29691291) Protein Fusion: out of frame {EWSR1:FLI1}	NA
EVT-P-0053879-T02-IM6-101	P-0053879-T02-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128665052	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128665052::chr22:g.29688409) Protein Fusion: in frame {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-053991-T01-IM6-102	P-0053991-T01-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128647321	5	197	True	197	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128647321::chr22:g.29684445) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-055969-T02-IM6-103	P-0055969-T02-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128674720	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128674720::chr22:g.29683823) Protein Fusion: in frame {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_effect_on_frame
EVT-P-0056323-T02-IM6-104	P-0056323-T02-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	out-of-frame	three_prime	128659413	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128659413::chr22:g.29685698) Protein Fusion: out of frame {EWSR1:FLI1}	NA
EVT-P-0056509-T02-IM6-105	P-0056509-T02-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	out-of-frame	three_prime	128648407	5	197	True	197	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128648407::chr22:g.29685275) Protein Fusion: out of frame {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0056693-T02-IM6-106	P-0056693-T02-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128654385	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128654385::chr22:g.29683723) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-0056712-T01-IM6-107	P-0056712-T01-IM6		Soft Tissue Sarcoma	ESST	EWSR1-FLI1	EWSR1	in-frame	three_prime	128672121	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128672121::chr22:g.29684489) Protein Fusion: in frame {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0057669-T01-IM6-108	P-0057669-T01-IM6		Bone Cancer	ES	EWSR1-FLI1	EWSR1	unknown	three_prime	128673876	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128673876::chr22:g.29688565) Protein Fusion: mid-exon {EWSR1:FLI1}	NA
EVT-P-0059143-T01-IM7-109	P-0059143-T01-IM7		Bone Cancer	ES	EWSR1-FLI1	EWSR1	unknown	three_prime	128660670	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128660670::chr22:g.29684654) Protein Fusion: mid-exon {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0059518-T02-IM7-110	P-0059518-T02-IM7		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128671917	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128671917::chr22:g.29684466) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-0060347-T02-IM7-111	P-0060347-T02-IM7		Bone Cancer	ES	EWSR1-FLI1	EWSR1	out-of-frame	three_prime	128648014	5	197	True	197	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128648014::chr22:g.29685134) Protein Fusion: out of frame {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site_effect_on_frame
EVT-P-0061850-T02-IM7-112	P-0061850-T02-IM7		Bone Cancer	ES	EWSR1-FLI1	EWSR1	out-of-frame	three_prime	128646072	5	197	True	197	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128646072::chr22:g.29685975) Protein Fusion: out of frame {EWSR1:FLI1}	NA
EVT-P-0062142-T03-IM7-113	P-0062142-T03-IM7		Bone Cancer	ES	EWSR1-FLI1	EWSR1	out-of-frame	three_prime	128644385	5	197	True	197	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128644385::chr22:g.29685696) Protein Fusion: out of frame {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0062142-T04-IM7-114	P-0062142-T04-IM7		Bone Cancer	ES	EWSR1-FLI1	EWSR1	out-of-frame	three_prime	128644385	5	197	True	197	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128644385::chr22:g.29685696) Protein Fusion: out of frame {EWSR1:FLI1}	NA
EVT-P-0063940-T02-IM7-115	P-0063940-T02-IM7		Bone Cancer	ES	EWSR1-FLI1	EWSR1	unknown	three_prime	128672287	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128672287::chr22:g.29688521) Protein Fusion: mid-exon {EWSR1:FLI1}	NA

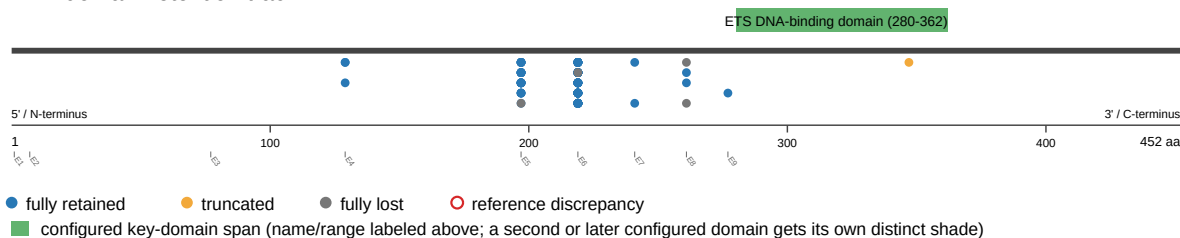
event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0065014-T01-IM7-116	P-0065014-T01-IM7		Bone Cancer	ES	EWSR1-FLI1	EWSR1	in-frame	three_prime	128666909	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128666909::chr22:g.29684574) Protein Fusion: in frame {EWSR1:FLI1}	NA
EVT-P-0065014-T02-IM7-117	P-0065014-T02-IM7		Soft Tissue Sarcoma	ESST	EWSR1-FLI1	EWSR1	in-frame	three_prime	128666909	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128666909::chr22:g.29684574) Protein Fusion: in frame {EWSR1:FLI1}	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0065672-T01-IM7-118	P-0065672-T01-IM7		Bone Cancer	ES	EWSR1-FLI1	EWSR1	out-of-frame	three_prime	128658571	6	219	True	219	True	exact match	retained	1.0	False	ETS DNA-binding domain			EWSR1 (NM_013986) - FLI1 (NM_002017) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128658571::chr22:g.29685397) Protein Fusion: out of frame {EWSR1:FLI1}	NA
EVT-P-001233-T01-IM3-119	P-0001233-T01-IM3		Bone Cancer	ES	EWSR1-FLI1	EWSR1	out-of-frame	five_prime	128646455	4	197	True	196	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False		ETS DNA-binding domain		EWSR1 (NM_005243) - FLI1 (NM_002017) Translocation: t(22;11)(q12.2;q24.3)(c.974+158_c.589+3575) Protein fusion: out of frame (FLI1-EWSR1)	NA

event_id	sample_id	patient_id	tumor_type	oncogene_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_croscheck_protein_position	fusion_annotation_croscheck_agrees	fusion_annotation_croscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0053737-T01-IM6-120	P-0053737-T01-IM6		Esophagegastic Cancer	EGC	FLI1-ERG	ERG	unknown	five_prime	128680564	9	347	False	346	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	disrupted	0.8192771084337349	True			ETS DNA-binding domain	FLI1 (NM_002017) - ERG (NM_182918) rearrangement: t(11;21)(q24.3;q22.2)(chr11:g.128680564::chr21:g.39755655) Protein Fusion: mid-exon {FLI1:ERG}	NA
EVT-P-0050027-T02-IM6-121	P-0050027-T02-IM6		Bone Cancer	ES	FLI1-EWSR1	EWSR1	in-frame	five_prime	128669484	5	219	True	218	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False		ETS DNA-binding domain		FLI1 (NM_002017) - EWSR1 (NM_013986) fusion: t(11;22)(q24.3;q12.2)(chr11:g.128669484::chr22:g.29683917) Protein Fusion: in frame {FLI1:EWSR1}	NA

domain retention outliers

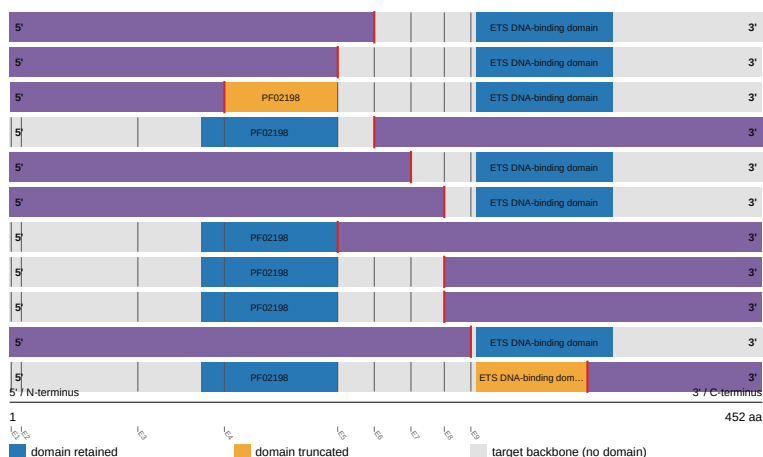
FLI1 domain-retention track



fusion schematic

FLI1 fusion-transcript schematic

Fusion order varies by row; follow each row's 5' and 3' labels (FLI1 portion is domain-colored, exon ticks). Partner-gene blocks share one color; row labels identify each partner.



Partner-block colors are decorative; row labels identify partners.

EWSR1 (x76) – ETS DNA-binding domain retained – multiple tumor types

EWSR1 (x26) – ETS DNA-binding domain retained – Bone Cancer, Salivary Gland Cancer, Soft Tissue

EWSR1 (x3) – ETS DNA-binding domain retained – Bone Cancer

EWSR1 (x3) – ETS DNA-binding domain lost – Bone Cancer

EWSR1 (x2) – ETS DNA-binding domain retained – Bone Cancer

EWSR1 (x2) – ETS DNA-binding domain retained – Bone Cancer

EWSR1 – ETS DNA-binding domain lost – Bone Cancer

ATM – ETS DNA-binding domain lost – Prostate Cancer

EWSR1 – ETS DNA-binding domain lost – Bone Cancer

EWSR1 – ETS DNA-binding domain retained – Bone Cancer

ERG – ETS DNA-binding domain truncated – Esophagogastric Cancer

■ breakpoint

reference comparison

Reference vs msk_impact_50k_2026

